

solution

→ Homogenous mixt of two or more components.

Solⁿ

Solute (st)

→ dissolve in another substance.
→ less quantity

Solvent (Solu)

→ In which substance gets dissolve
→ more quantity

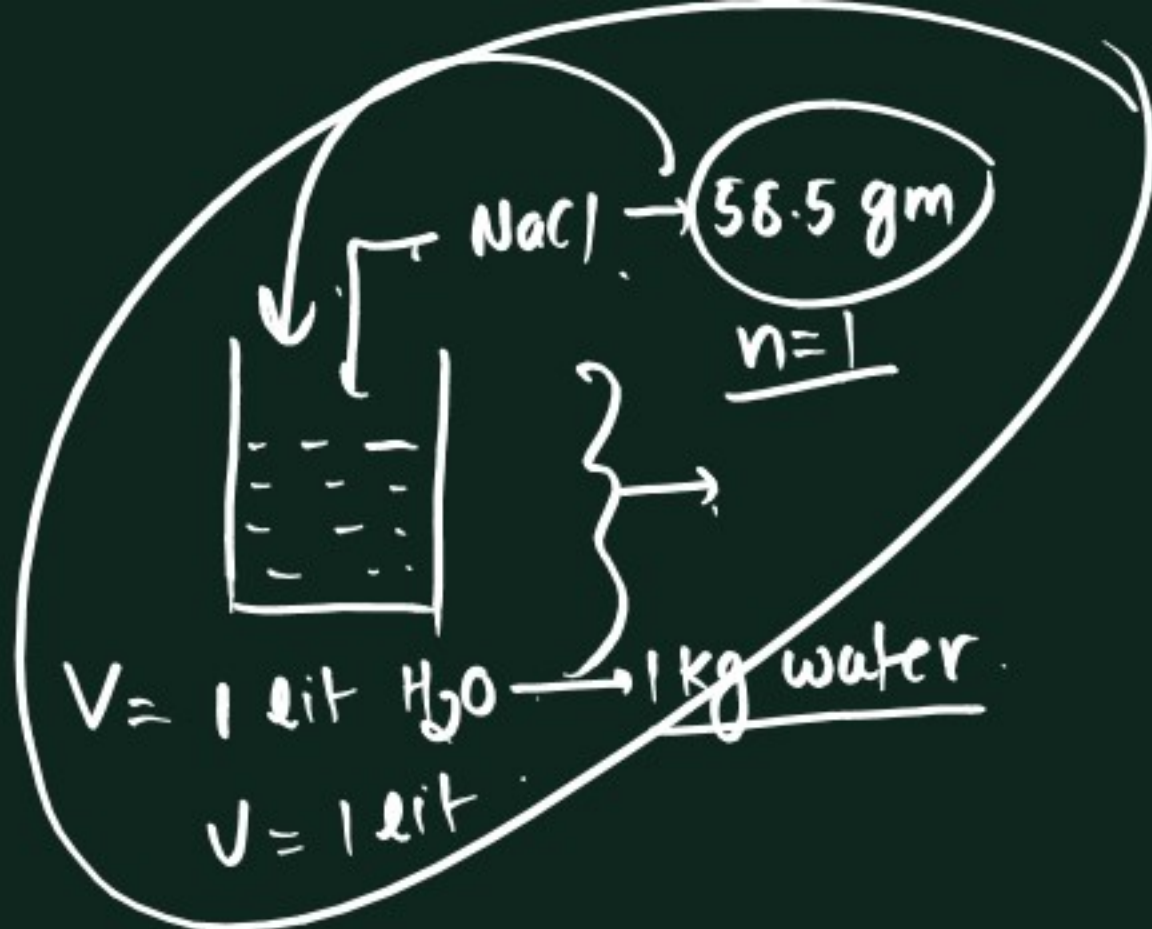
Comp-1 + Comp-2

Solid
solute

liq
solvent

Solⁿ

liq



conc. terms →

- 1) % by mass or $\frac{m}{m}$
- 2) % by strength or $\frac{m}{V}$
- 3) % by vol^m or V/V
- 4) mole fraction (x)
- 5) ppm

- 6) strength
- 7) molarity ✓
- 8) molality

① % by mass or $\frac{m}{m}\%$ or $\frac{w}{w}\%$ $\rightarrow$

$$\% \frac{m}{m} = \frac{\text{mass of solute}}{\text{mass of soln}} \times 100$$

$$\% \frac{m}{m} = \frac{m_{st}}{m_{sol}} \times 100$$

e.g. aqueous solⁿ containing 40% by mass of HCl $\rightarrow$
 $\rightarrow$ 100 gm of solⁿ $\rightarrow$ 40 gm of HCl

② % by strength or $\frac{m}{v}\%$

$$\% \frac{m}{v} = \frac{\text{mass of solute}}{\text{vol^m of solⁿ} \times 100 = \frac{m_{st}}{V_{soln}} \times 100$$

e.g. aq. solⁿ containing 40% $\frac{m}{v}$ of HCl
 $\rightarrow$ 100 ml of solⁿ $\rightarrow$ 40 gm of HCl

3) % by vol^m or $\frac{v}{v}\%$ $\rightarrow$

$$\textcircled{2} \quad 30 = \frac{x}{x+500} \times 100$$

$$x = 214 \text{ gm}$$

$$d = \frac{m}{V}$$
$$V_{sol} = \frac{m_{sol}}{d_{sol}} = \frac{500}{1.2}$$

$$\textcircled{3} \quad \% \text{ by strength} = \frac{m_{st}}{V_{soln}} \times 100$$

$$= \frac{30}{V_{soln}} \times 100$$

$$= \frac{30^6}{500} \times 1.2 \times 100$$
$$= 7.2 \%$$

$$4) d_{\text{sol}} = 1.5 \frac{\text{gm}}{\text{ml}}$$

$$d = \frac{m}{V}$$

$$m = d \times V \\ = 1.5 \times 100 \\ = 150$$

Sum of moles fraction = 1

$$x_{\text{st}} + x_{\text{solv}} = 1$$

$$\frac{n_{\text{st}}}{n_{\text{sol}}} + \frac{n_{\text{solv}}}{n_{\text{sol}}} = 1$$

$$\frac{n_{\text{st}} + n_{\text{solv}}}{n_{\text{sol}}} = 1$$

$$\rightarrow \text{by } \frac{m}{V} = 30\%$$

$$100 \text{ ml sol}^n \rightarrow 30 \text{ gm NaCl}$$

$$\text{by mass} = 20\%$$

④ Mole fraction (x) →

$$x_{\text{st}} = \frac{n_{\text{st}}}{n_{\text{sol}}} = \frac{n_{\text{st}}}{n_{\text{st}} + n_{\text{solv}}}$$

$$x_{\text{solv}} = \frac{n_{\text{solv}}}{n_{\text{st}} + n_{\text{solv}}}$$

$$\frac{x_{\text{st}}}{x_{\text{solv}}} = \frac{n_{\text{st}}}{n_{\text{solv}}}$$

$$n_{\text{glucose}} = \frac{30}{180} = \frac{1}{6}$$

$$n_{\text{H}_2\text{O}} = \frac{90}{18} = 5$$

$$x_{\text{glucose}} = \frac{\frac{1}{6}}{\frac{1}{6} + 5} = \frac{1}{31}$$



34.2% by mass

$$100 \text{ gm sol}^n \rightarrow 34.2 \text{ gm sucrose}$$

$$m_{\text{H}_2\text{O}} = 100 - 34.2 \\ = \underline{65.8 \text{ gm}}$$

$$n_{\text{sucrose}} = \frac{34.2}{342} = 0.1$$

$$n_{\text{H}_2\text{O}} = \frac{65.8}{18} = 3.6$$

$$x_{\text{sucrose}} = \frac{0.1}{0.1 + 3.6} \\ = \frac{1}{37} = \frac{1}{37}$$

$$\text{PPM} = \frac{m_{\text{st}}}{m_{\text{sol}^n}} \times 10^6$$

parts per million

Strength (S)

$$S = \frac{\text{mass of solute} \checkmark}{\text{vol}^m \text{ of sol}^n \text{ in lit} \checkmark}$$

$$S = \frac{m_{st}}{V_{sol} \text{ in lit.}}$$

$$S = \frac{m_{st}}{V_{sol} \text{ in ml}} \times 1000$$

$$\text{unit} \rightarrow \frac{\text{gm}}{\text{lit.}}$$

Molarity (M) or conc (C)

$$M = \frac{\text{no. of moles of solute}}{\text{vol}^m \text{ of sol}^n \text{ in lit.}}$$

$$M = \frac{n_{st}}{V_{sol} \text{ in lit.}}$$

$$M = \frac{n_{st}}{V_{sol} \text{ in ml}} \times 1000$$

$$\text{unit} \rightarrow \frac{\text{mol}}{\text{lit}} \text{ or } M \text{ or Molar.}$$

$$S = \frac{5.6 \times 1000}{500} = 11.2 \frac{\text{gm}}{\text{lit.}}$$

$$M = \frac{0.1}{500} \times 1000 \\ = \underline{0.2 M}$$

$$n_{st} = \frac{5.6}{56} = 0.1$$

$$M = \frac{n_{st} \times 1000}{V_{soln}}$$

$$0.2 = \frac{n_{st}}{200} \times 1000$$

$$n_{st} = 0.08$$

$$m_{glucose} = 0.08 \times 180 \\ = \underline{14.4 \text{ gm}}$$

$$g) d_{soln} = 1.5 \frac{\text{gm}}{\text{ml}}$$

$$100 \text{ gm soln} \rightarrow \underline{30 \text{ gm NaCl}} \quad n_{st} = \frac{30}{58.5}$$

$$d = \frac{m}{V}$$

$$V = \frac{m}{d} = \frac{100}{1.5} \text{ ml}$$

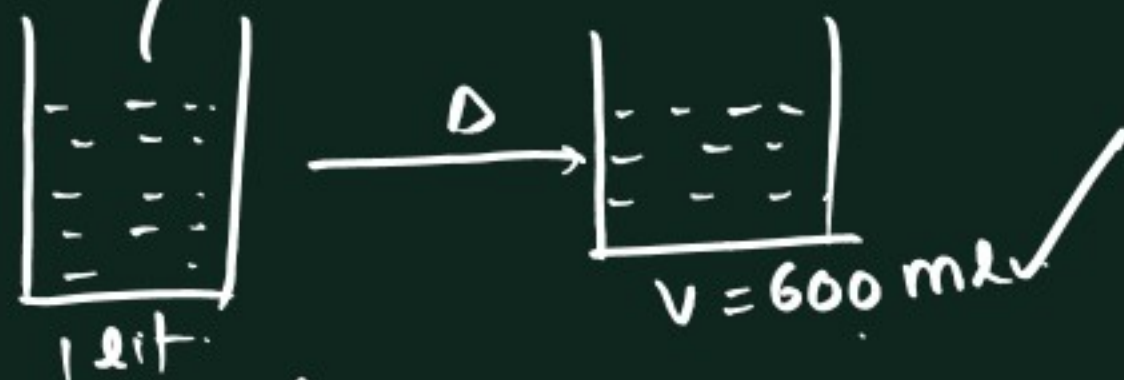
$$M = \frac{\frac{30/58.5}{100} \times 1000}{1.5} = \underline{7.6 \text{ M}}$$

$$\textcircled{4} \quad H_2O = 1 \text{ lit} = 1 \text{ kg} = 1000 \text{ gm} \quad n_{H_2O} = \frac{1000}{18} = 55.55$$

$$M = \frac{55.55}{1}$$

1 lit 0.5 M HCl 2 3.25 gm

3.25 gm $\rightarrow (n_{\text{HCl}})_{\text{remove}} = \frac{3.25}{36.5}$



$M = 0.5$ ✓

$n_{\text{HCl}} =$
 $M = \frac{n}{V}$

$n = M \times V$
 $= 0.5 \times 1$
 $= 0.5$

$n_{\text{HCl}} = 0.5 - \frac{3.25}{36.5}$ ✓

$M = 0.68 \text{ M}$

* Determination of moles and millimoles $\rightarrow$
(mmoles)

$$M = \frac{n_{st}}{V}$$

$$n_{st} = M \times V (\text{lit})$$

(moles)

$$n_{st} = M \times V (\text{ml})$$

(mmole)

$$\text{millimoles} = \text{moles} \times 10^{-3}$$

$$M = \frac{n_{st} (\text{moles})}{V (\text{lit})}$$

$$M = \frac{n_{st} (\text{mmoles})}{V (\text{ml})}$$

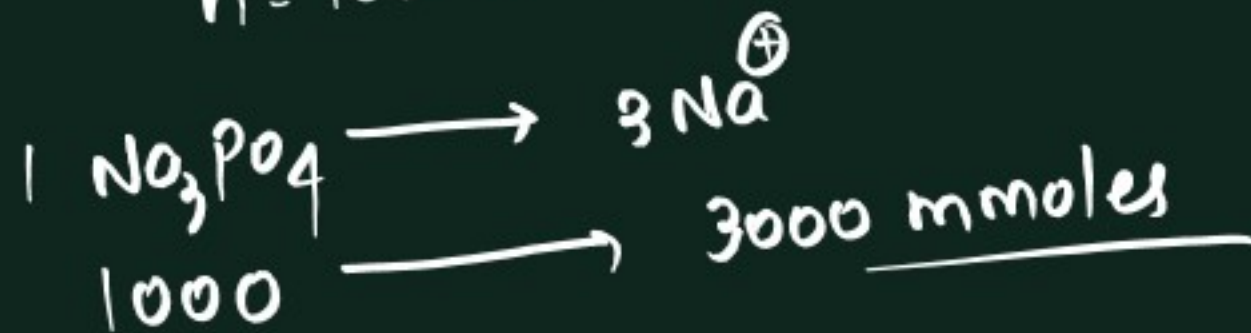


$$M = 2$$

$$V = 500 \text{ ml}$$

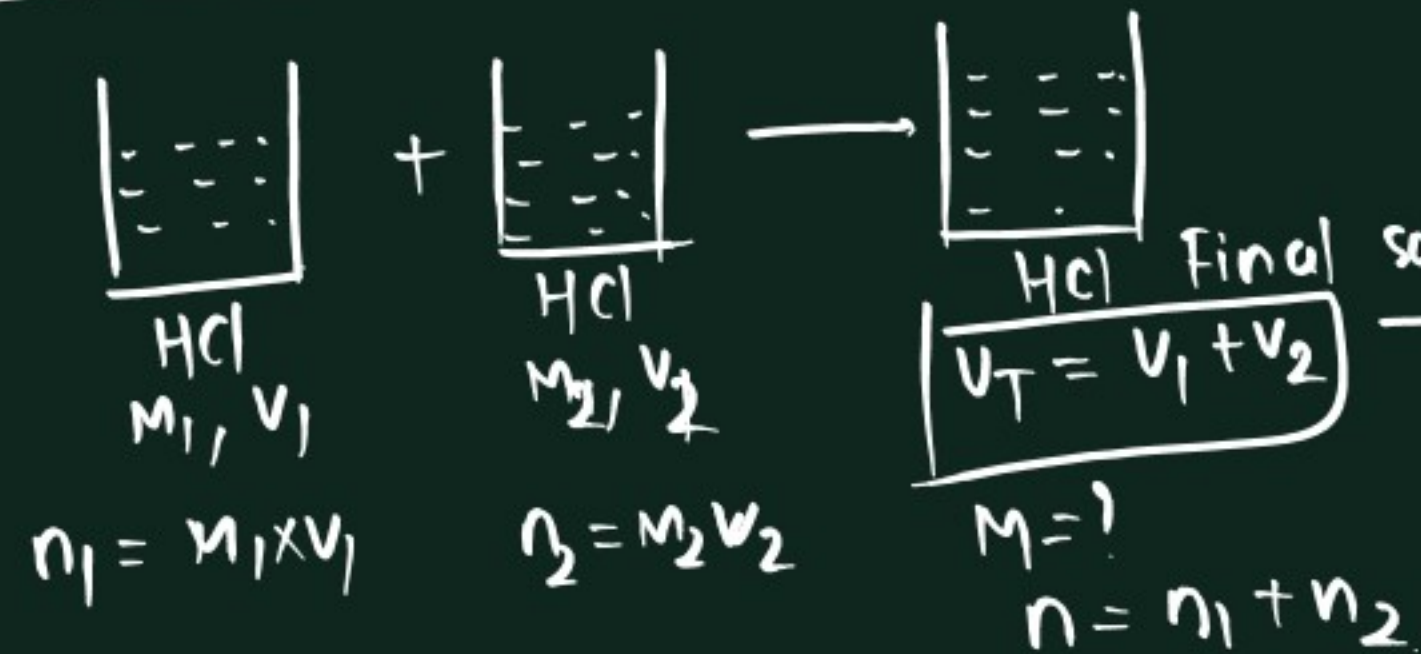
$$n_{\text{mmoles}} = M \times V (\text{ml})$$
$$= 2 \times 500$$

$$n = 1000 \text{ mmole} \checkmark$$



$$= 25 \times 10^{-3} \text{ mole}$$

* Mixing of same type of solⁿ →



$$M = \frac{n_1 + n_2}{V_T} = \frac{M_1 V_1 + M_2 V_2}{V_1 + V_2}$$

$$M = \frac{M_1 V_1 + M_2 V_2}{V_T}$$

solⁿ after mixing
when final vol^m is
not given

$\frac{M}{2}$, 500ml
200ml, decimolar solⁿ
 $\left(\frac{1}{10}\right)$

$$\begin{aligned}
 M &= \frac{M_1 V_1 + M_2 V_2}{V_T} = \frac{\frac{1}{2} \times 500 + \frac{1}{10} \times 200}{1000} \\
 &=
 \end{aligned}$$

$$n_{\text{NaCl}} = 0.2 \times 200 = 40 \text{ mmole}$$

n_{Cl^-}

$$n_{\text{Cl}^-} = 40 \text{ mmole}$$

$$n_{\text{CaCl}_2} = 0.2 \times 400 = 80 \text{ mmole}$$

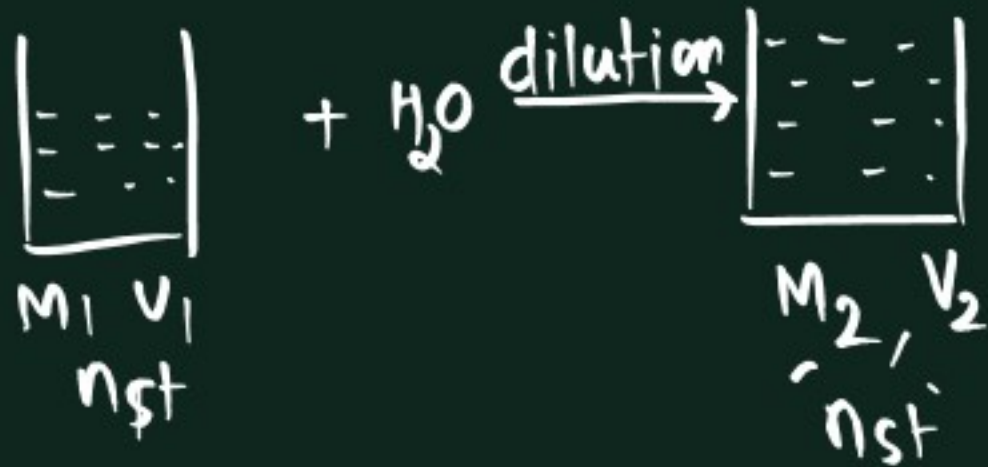
$$n_{\text{Cl}^-} = 160 \text{ mmole}$$

$$n_{\text{Cl}^-} = 200 \text{ mmole}$$

$$M_{\text{Cl}^-} = \frac{n_{\text{Cl}^-}}{V} = \frac{200 \text{ mmole}}{600 \text{ ml}} = \frac{1}{3} \text{ M}$$

~~Applica~~
Effect of dilution

Dilution $\rightarrow$ addition of water



before dilution = after dilution
 $n_{st} = n_{st}$

$$M_1 \times V_1 = M_2 V_2$$

By dilution $\Rightarrow V \uparrow \Rightarrow M \downarrow$

$$\underline{M_1 V_1} = \underline{M_2 V_2} \quad V_1 = \underline{500 \text{ ml}}$$

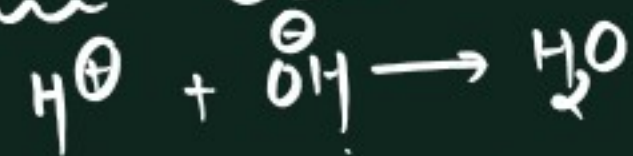
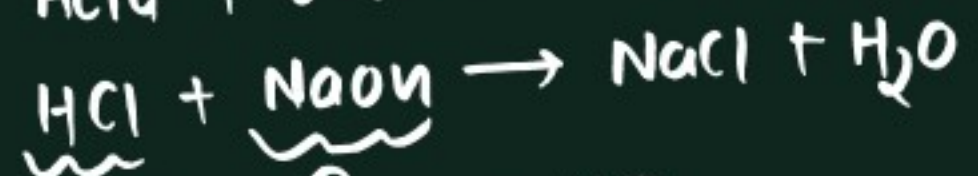
$$V_2 = \underline{1250 \text{ ml}}$$

water added
 $= V_2 - V_1$
 $= \underline{750 \text{ ml}}$



* Application in neutralization rxn $\rightarrow$

Acid + Base $\rightarrow$ salt + water



complete neutralization $\rightarrow$ The $\text{H}^{\oplus}$ ion from acid are completely neutralized by base and vice versa.

$\rightarrow$ final soln is neutral.

$$\Rightarrow \boxed{n_{\text{H}^{\oplus}} = n_{\text{OH}^{\ominus}}}$$

Incomplete neutralization $\rightarrow$ either $\text{H}^{\oplus}$ or $\text{OH}^{\ominus}$ are in excess amt.

$\rightarrow$ final soln $\rightarrow \text{H}^{\oplus} > \text{OH}^{\ominus} \rightarrow$ acidic soln

$\text{H}^{\oplus} < \text{OH}^{\ominus} \rightarrow$ basic soln

Acidity → define for Base
→ no. of replacable $\text{OH}^\ominus$ ion present in base

e.g

1) monoacidic base → $(1 \text{ OH}^\ominus)$ → NaOH , KOH , RbOH , CsOH , NH_4OH etc

2) diacidic base → $(2 \text{ OH}^\ominus)$ → Ca(OH)_2 , Mg(OH)_2 , Fe(OH)_2

3) triacidic base → $(3 \text{ OH}^\ominus)$ → Al(OH)_3 , Fe(OH)_3

Basicity → define for acid.
→ no. of replacable $\text{H}^\oplus$ ion in acid

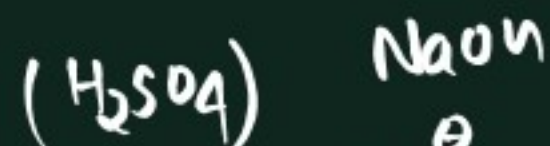
1) monobasic acid → HCl , HBr , HNO_3 , HI , etc CH_3COOH

2) dibasic acid → H_2SO_4 , H_2CO_3 , H_3PO_3

3) tribasic acid → H_3PO_4

$$n_{\text{OH}^\ominus} = \text{acidity} \times M \times V$$

$$n_{\text{H}^\oplus} = \text{basicity} \times M \times V$$



$$n_{\text{H}^\oplus} = n_{\text{OH}^\ominus}$$

$$2 \times M \times V = 1 \times M \times V$$

$$2 \times 0.2 \times V = 1 \times 0.1 \times 500$$

$$V = \frac{50}{0.4}$$

$$= 125 \text{ ml}$$

$$\text{HCl} \rightarrow M = 0.2$$

$$V = 400 \text{ ml}$$

$$n_{\text{H}^+} = 1 \times 0.2 \times 400$$

$$= 80 \text{ mmole}$$

After reaction $\rightarrow$

$$n_{\text{OH}^-} = 240 \text{ mmole}$$

$$M_{\text{OH}^-} = \frac{n_{\text{st}}}{V} = \frac{240 \text{ mmole}}{1200} = \frac{24}{120} = 0.2 \text{ M}$$

$$\text{Ca(OH)}_2 \rightarrow M = 0.2$$

$$V = 800 \text{ ml}$$

$$n_{\text{OH}^-} = 2 \times 0.2 \times 800$$

$$= 320$$

* $n_{\text{H}^+} = 2 \times 0.2 \times 400 = 80 \text{ mmole}$

$n_{\text{OH}^-} = 1 \times 0.2 \times 500 = 100 \text{ mmole}$

After reaction $\rightarrow n_{\text{OH}^-} = 20$

$$M_{\text{OH}^-} = \frac{20}{100} =$$

Molality (m) $\rightarrow$

$$m = \frac{\text{no. of moles of solute}}{\text{mass of solvent in kg}}$$

$$m = \frac{n_{\text{st}}}{m_{\text{solv in kg}}}$$

$$m = \frac{n_{\text{st}}}{m_{\text{solv in gm}}} \times 1000$$

unit $\rightarrow \frac{\text{mol}}{\text{kg}}$
or m
or mola

Temp dependent $\rightarrow$ % by strength, M, S
% by vol^m

(contain vol^m)

Temp independent $\rightarrow$ % by mass, x
molality, PPM

$$1) \quad 100 \text{ gm sol}^n \longrightarrow 9.8 \text{ gm of } \text{H}_2\text{SO}_4 \longrightarrow n_{\text{st}} = 0.1$$

$$m_{\text{solvent}} = 100 - 9.8 \\ = 90.2 \text{ gm}$$

$$m = \frac{0.1}{90.2} \times 1000 = \frac{100}{90.2}$$

$$\boxed{m = 1.1 \frac{\text{mol}}{\text{kg}}}$$

$$2) \quad 93\% \frac{m}{V}$$

$$100 \text{ ml sol}^n$$

$$93 \text{ gm of } \text{H}_2\text{SO}_4$$

$$\longrightarrow n_{\text{H}_2\text{SO}_4} = \frac{93}{98}$$

$$d_{\text{sol}^n} = \frac{m_{\text{sol}}}{V_{\text{sol}^n}}$$

$$m_{\text{sol}} = d \times V \\ = 1.98 \times 100 = 198 \text{ gm}$$

$$m_{\text{sol}^n} = 198 - 93 \\ = 105 \text{ gr}$$

$$m = \frac{93/98 \times 1000}{105}$$

$$= \underline{9.03 \text{ molal}}$$

$$m = 1$$

$$m_{sol}^n = 1000 \text{ gm}$$

$$m_{st} = x$$

$$m_{sol}^v = 1000 - x$$

$$I = \frac{n_{st}}{m_{sol}^v} \times 1000 \text{ in gm}$$

$$I = \frac{\frac{x}{180}}{1000 - x} \times 1000$$

$$x = \cancel{219 \text{ gm}} \quad 152 \text{ gm}$$

Relation b/w diff. conc. terms.

② $\therefore$ by mass & $M \rightarrow$

$$M = \frac{n_{st}}{V_{sol}^n} \times 1000 \text{ in ml}$$

$$= \frac{m_{st} \times 1000}{V_{sol}^n \times M \cdot w_{st}}$$

$$= \frac{m_{st} \times d_{sol}^n \times 1000 \times 10}{m_{sol}^n \times M \cdot w_{st}}$$

$$M = \frac{\therefore \text{by mass} \times d \times 10}{M \cdot w_{st}}$$

① $S \& M$

$$S = \frac{m_{st}}{V_{sol}^n}$$

$$M = \frac{n_{st}}{V_{sol}^n} = \frac{m_{st}}{M \cdot w_{st} \times V_{sol}^n}$$

$$M = \frac{S}{M \cdot w_{st}}$$

$$d_{sol}^n = \frac{m_{sol}^n}{V_{sol}^n}$$

$$V_{sol}^n = \frac{m_{sol}^n}{d_{sol}^n}$$

3) molality and mole fraction →

$$\frac{\chi_{st}}{\chi_{solu}} = \frac{n_{st}}{n_{solu}}$$
$$= \frac{n_{st} \times M \cdot w_{solu}}{m_{solu}} \times \frac{1000}{1000}$$

$$\frac{\chi_{st}}{\chi_{solu}} = \frac{m \times M \cdot w_{solu}}{1000}$$

④ $m \& M \rightarrow$

$$d_{soln} = M \left[\frac{M \cdot w_{st}}{1000} + \frac{1}{m} \right]$$

$$m = \frac{1000 \times M}{1000 \times d - M \cdot w_{st} \times M}$$

$$m = 0.98$$

